

Class 06 — Homework

Bong Study Hub — Foundation Batch 2026 · Class 06: Variables & Data Types

Giving Values a Label That Can Change

Estimated time: 30–45 minutes **Total:** 30 marks · **Optional Challenge:** up to 3 bonus marks **Tools needed:** pen and paper, or a simple document. A computer with Python is helpful for checking your work, but not required to answer any question. **Prerequisite:** Classes 05 and 06. No arithmetic operators, `input()`, loops, or conditionals are needed anywhere in this assignment.

What This Assignment Is About

In class, we answered a question Class 05 left open: how does a program hold onto something that changes? We met the **variable** — a named, changeable place to store a value — created variables with `=`, learned that `=` means "store" and not mathematical equality, saw why `print(name)` and `print("name")` produce completely different output, watched reassignment completely replace an old value, and met four data types: `int`, `float`, `str`, and `bool`.

This assignment asks you to **read and trace code, not just recall facts about it**. Several questions give you a snippet of Python and ask what it does, exactly like the write-run-read-fix loop from Class 05, just on paper.

Instructions

- **Answer in your own words** where a question asks for an explanation.
 - For "predict the output" or "trace the code" questions, write the **exact** text Python would display, in order.
 - **No arithmetic operators, `input()`, conditionals, or loops appear anywhere in this homework** — if your answer to any question involves one of those, simplify it back down to variables, assignment, and `print()`.
 - You may refer to your **Student Notes** while answering.
 - If you're unsure, **explain your reasoning anyway** rather than leaving a blank — partial thinking is worth more than an empty space.
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Section A — Core Concepts

Question 1 — Why Variables? (2 marks)

In your own words, explain **why Class 05's fixed, typed-once text wasn't enough for something like a scoreboard that changes during a game.**

Question 2 — What Is a Variable? (2 marks)

In your own words, **explain what a variable is**, using the labeled-box idea from class if it helps.

Question 3 — Assignment vs. Equals (2 marks)

Explain why `=` in Python is not the same as the `=` you use in math class.

Question 4 — Variables in print() (2 marks)

Explain the difference between what `print(city)` and `print("city")` would each display, assuming `city` is a variable that holds the value `"Kolkata"`.

Question 5 — The Four Data Types (2 marks)

Name the four data types covered in class, and give one example value of each.

Section B — Reading and Reasoning About Code

Question 6 — Predict the Output (3 marks)

```
name = "Rohan"  
score = 42  
print("Player:", name)  
print("Score:", score)
```

Write the exact output, line by line, in the order it would appear.

Question 7 — Find the Bug (3 marks)

A student wanted to print the value stored in `city`, but got this output instead of the city's name:

```
city = "Kolkata"  
print("city")
```

Output:

```
city
```

What's wrong with the second line? Rewrite it so it correctly displays the value stored in `city`.

Question 8 — Reassignment Trace (3 marks)

```
temperature = 20  
print(temperature)  
temperature = 25  
print(temperature)  
temperature = 18  
print(temperature)
```

Write the exact output, in order, and explain in one sentence what happened to each earlier value once the next line reassigned `temperature`.

Question 9 — Identify the Data Type (3 marks)

For each value below, write whether it is an `int`, a `float`, a `str`, or a `bool`:

Value	Data type
<code>`42`</code>	
<code>`3.14`</code>	
<code>""hello""</code>	
<code>`True`</code>	
<code>""True""</code>	
<code>`0.0`</code>	

Question 10 — Fix the Variable Name (3 marks)

Each of these variable names has a problem. Identify what's wrong with each one, and rewrite it as a valid variable name:

```
1st_place = "Aisha"
student age = 20
class = "Foundation Batch"
```

Section C — Class Connection & Your Own Program

Question 11 — Connecting Class 04 and Class 06 (2 marks)

Fill in the blank below, using ideas from today's class:

INFORMATION

↓

REPRESENTATION (Class 04)

↓

BITS (Class 04)

↓

_____ (Class 06)

Briefly explain, in one sentence, **what a variable gives you that raw bits don't**.

Question 12 — Write Your Own Variable Profile Card (3 marks)

Write a short program (on paper is fine) with **at least four variables** — one string, one integer, one float, and one boolean — describing a person, animal, or character of your choice. Then write at least two ``print()`` lines that combine literal text and variables using commas.

Optional Challenge — Bonus (up to 3 marks)

This section is entirely optional and will not affect your base grade.

Trace the program below very carefully — it reassigns some variables and leaves others untouched.

Write the exact output, in order.

```
name = "Meera"
score = 10
is_winner = False

print(name, "starts with a score of", score)

score = 10
score = 30
is_winner = True

print(name, "now has a score of", score)
print("Is", name, "the winner?", is_winner)

name = "Meera R."
print(name, "is the final name on record.")
```

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document.
- **Option C:** Submit as a simple PDF.

Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 12 questions.
- ☐ For every "predict the output" or "trace the code" question, I wrote the exact text, in the exact order.
- ☐ Where unsure, I explained my reasoning instead of leaving a blank.
- ☐ I did not use arithmetic operators, `input()`, conditionals, or loops anywhere.

☐ I reviewed my work before submitting.